Construction of a Dataset on Italian Medicines

A PROJECT REPORT

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1. Sources Used

- **AIFA (Agenzia Italiana del Farmaco) official website:** For obtaining PDFs of the Riassunto delle Caratteristiche del Prodotto (RCP).
- **Python Libraries:** Selenium, Fritz (PyMuPDF), Pandas, Re, Tqdm: Used for scraping and processing documents.
- **Power BI:** For visualizing key aspects of the dataset (active ingredients, therapeutic indications, side effects).
- **SQLite & DB Browser for SQLite:** Used for database normalization and analytical queries.

2. Methodology Adopted

Pre-Step: Preparing the Submerged Dataset

- Datasets A and H were first downloaded.
- Empty rows were removed from both datasets.
- Rows 2464 to 2795 (inclusive) were selected to form a new, clean Submerged Dataset.

Step 1: Collection of AIC Codes and PDFs:

- A list of over 300 AIC codes was compiled.
- Selenium automation was implemented to search the AIFA site using these AICs.
- The RCP PDFs were downloaded from each medicine's product page.

Step 2: Text Extraction from PDFs

- PyMuPDF (fitz) was used to extract text from each PDF.
- Only the following sections were retained:
- 4.1 Therapeutic indications
- 4.2 Posology and method of administration
- 4.3 Contraindications
- 4.4 Special warnings and precautions for use
- 4.5 Interactions with other medicinal products
- 4.6 Fertility, pregnancy and lactation
- 4.7 Effects on ability to drive and use machines
- 4.8 Undesirable effects
- 4.9 Overdose
- 6.2 Incompatibilities
- Extraction logic ensured each section stopped before the next number (e.g., stop 4.9 before 5.0).

Step 3: Final Dataset Construction

- Extracted data was saved into a structured Excel file:FINAL_UPDATED_XLSX.xlsx.
- Power BI was used to build visualizations:
- Distribution by Active Ingredient
- Therapeutic Indication Frequency
- Side Effect Prevalence

Step 4: Database Design and Normalization

- DB Browser for SQLite was used to create the databaseaifa_medicines.db.
- Normalization was done into multiple relational tables:
- medicines (main data)
- atc_classifications
- contraindications
- side_effects
- Foreign keys were applied where needed.

Step 5: Analytical SQL Queries

Five SQL queries were created to generate insights:

- Count of medicines grouped by ATC code.
- Most frequent therapeutic indications.
- Most common contraindications.
- Medicines with most side effects.
- Medicines in each equivalence group.

3. Choices Made

- **Sections extracted:** Based on clinical relevance and the requirements outlined in the project document, all key sections were targeted, specifically sections 4.1 to 4.9 and 6.2.

- **PDF filtering:** No files were skipped. Even those without clearly extractable sections were manually reviewed and included to ensure completeness.

- **Side Effects:** All relevant sections, from 4.1 to 4.9 and 6.2, were extracted without exception.

- **Text handling:** Paragraphs were cleaned, and all required data was included. Nothing was omitted or left out.

- **Normalization:** Tables were logically split, and relationships were defined to reflect dependencies—such as linking ATC classifications to their respective medicines.

4. Difficulties Encountered

- **PDF format inconsistency**: Some PDFs lacked consistent section formatting, which initially caused partial extraction errors. To resolve this, all PDFs were converted to .txt files before performing the extraction process.

- **OCR issues**: A few documents had non-text-renderable content. These were also converted to .txt files, which enabled successful text-based extraction.

- **Power BI usage**: There was some initial difficulty due to limited prior experience with Power BI. However, after a short learning period, the required visualizations were implemented successfully without further issues.

5. Optional Code and Scripts

All automation and extraction were done using custom Python scripts. The code is available and can be shared on request for further validation.

Final Note

The project effectively combined data scraping, cleaning, visualization, and database normalization. It simulated a real-world data engineering pipeline in a healthcare context.